

# SOC Incident Investigation Report - Scenario 02

## Alert Name

Suspicious SSH Activity with Persistence Establishment

## Severity

Critical

## Data Source

Linux secure.log

## Executive Summary

A brute-force attempt from IP 203.0.113.50 was observed but did not result in successful authentication. A separate IP address, 198.51.100.25, successfully authenticated using the deploy account, escalated privileges to root, stopped the Apache service, obtained a root shell, created a new user account (backup2), changed its password, and successfully logged in using the newly created account. These activities indicate persistence establishment and possible system compromise.

## Timeline

01:10:15 - Successful login by backup from 10.0.0.5  
01:12:30 - Scheduled backup job executed by root  
02:15:01 - Failed SSH login (root) from 203.0.113.50  
02:15:05 - Failed SSH login (root) from 203.0.113.50  
02:15:09 - Failed SSH login (admin) from 203.0.113.50  
02:15:15 - Failed SSH login (test) from 203.0.113.50  
02:18:44 - Successful login by deploy from 198.51.100.25  
02:20:10 - deploy used sudo to stop Apache service  
02:20:15 - deploy obtained interactive root shell  
02:25:40 - New user account backup2 created  
02:26:01 - Password set for backup2  
02:30:12 - Successful login by backup2 from 198.51.100.25

## Indicators of Compromise (IOCs)

Suspicious IP: 198.51.100.25  
New User: backup2  
Compromised Account: deploy  
Service Manipulation: apache2 stopped  
Privilege Escalation: sudo to root shell

## MITRE ATT&CK; Mapping

T1078 - Valid Accounts  
T1548 - Abuse Elevation Control Mechanism

T1059 - Command and Scripting Interpreter

T1136 - Create Account

T1489 - Service Stop

TA0003 - Persistence

TA0004 - Privilege Escalation

## **Impact**

An attacker or unauthorized user appears to have obtained elevated privileges, created a secondary account for persistence, and verified access through the newly created account. Service disruption was also observed through the stopping of Apache.

## **Containment Recommendations**

- Isolate the affected host
- Disable deploy and backup2 accounts
- Reset credentials and review authentication logs
- Block suspicious IP 198.51.100.25
- Preserve logs and collect forensic evidence
- Review cron jobs, SSH keys, and persistence mechanisms
- Verify integrity of critical services and applications

## **Lessons Learned**

Analysts should focus on successful compromise activity rather than only failed login attempts. The brute-force activity acted as a distraction, while the actual impact originated from a different IP address that successfully authenticated and established persistence.